

FAULT CODE 351 - Injector Power Supply - Bad Intelligent Device or Component

Warnings and Cautions

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

CAUTION

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test lead when taking a measurement: Part Number 3164133 - male Deutsch™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Read the fault codes.	Fault Code 441 active or inactive?
	STEP 1B. Operate the engine.	Will the engine start?
	STEP 1C. Read the fault codes.	Fault Code 351 active?
STEP 2.	Check the ECM power supply.	
	STEP 2A. Inspect the ECM power supply connectors and fuses.	Dirty or damaged pins?
	STEP 2B. Check for an open circuit.	Less than 0.5 ohms?

STEPS		SPECIFICATIONS
	STEP 2C. Check for an open circuit in the ECM power supply circuit.	Less than 10 ohms?
STEP 3.	Validate the occurrence of this fault code.	
	STEP 3A. Operate the engine and determine if fault code condition exists.	Fault Code 351 occurs during engine operation?
STEP 4.	Check ECM calibration and clear fault codes.	
	STEP 4A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 4B. Disable the fault code.	Fault code inactive?

STEP 1. Check the fault codes.

STEP 1A. Read the fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Read the fault codes. • Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 441 active or inactive? YES Repair : Troubleshoot Fault Code 441 first.	Appropriate fault code troubleshooting tree
	Fault Code 441 active or inactive? NO	1B

STEP 1B. Operate the engine.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Operate the engine.	Will the engine start? YES	1C
	Will the engine start? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	4A

STEP 1C. Read the fault codes.**Conditions :**

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Operate the engine and determine whether the fault code condition still exists. • Operate the engine at high idle, no load. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 351 active? YES Repair : A marginal battery voltage condition is possible. Make sure the batteries are fully charged or replace, if needed. It is likely that a marginal battery condition will result in Fault Code 351 becoming active at engine cranking.	2A
	Fault Code 351 active? NO	4A

STEP 2. Check the ECM power supply.

STEP 2A. Inspect the ECM power supply connectors and fuses.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ECM power supply harness from the ECM connector.

Action	Specification/Repair	Next Step
Inspect the ECM power supply and fuses for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the sensor or harness connector. Clean the connector and pins. Repair or replace the damaged harness, pins, fuses, or connectors. Refer to Procedure 019-043 in the Associated Procedures Table.	4A
	Dirty or damaged pins? NO	2B

STEP 2B. Check for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ECM power supply harness from the ECM connector.

Action	Specification/Repair	Next Step
Check for an open circuit. • Measure the resistance between the ECM battery SUPPLY (+) pins at the power connector of the ECM power supply harness and the battery positive (+) pins at the battery positive (+) connection. Use the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 0.5 ohms? YES	2C
	Less than 0.5 ohms? NO Repair : Repair or replace the ECM power supply harness, fuses, or fuse holders. Clean the battery terminal connections. See equipment manufacturer service information.	4A

STEP 2C. Check for an open circuit in the ECM power supply circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ECM power supply harness from the ECM connector.

Action	Specification/Repair	Next Step

Check for an open circuit in the ECM ground circuit. Measure the resistance between the battery negative (-) pins at the ECM power supply connector to engine block ground. Use the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	3A
	Less than 10 ohms? NO Repair : Repair or replace the ECM power supply harness, fuses, or fuse holders. Clean the battery terminal connections. See equipment manufacturer service information.	4A

STEP 3. Validate the occurrence of this fault code.

STEP 3A. Operate the engine and determine if fault code condition exists.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Operate the engine and determine whether the fault code condition still exists. • Operate the engine at high idle, no load. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 351 occurs during engine operation? YES Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	4A
	Fault Code 351 occurs during engine operation? NO Repair : A marginal battery voltage condition is possible. Make sure the batteries are fully charged or replace, if needed. It is likely that a marginal battery condition will result in Fault Code 351 becoming active at engine cranking.	4A

STEP 4. Check ECM calibration and clear fault codes.

STEP 4A. Check if an ECM calibration update is available.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Compare the ECM code and revision in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision numbers are found in the Calibration Information section of System ID and Dataplate in Features & Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	4B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)	4B

STEP 4B. Disable the fault code.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault Code inactive? YES	Repair complete
	Fault Code inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/218/218-019-031.html)	QSB4.5 CM2150 B108	4326163
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/216/216-019-031.html)	QSB6.7 CM2150 B109	4326168
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F104	4332741
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F108	4332746
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/331/331-019-031.html)	QSB7 CM2880 B117	4358390
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/209/209-019-031-tr.html)	ISB3.9 CM2200 B107	4310792
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/164/164-019-031.html)	ISL9.5 CM2150 SN	4310608

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Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/206/206-019-031-tr.html)	QSNT14 CM876 N102	4325993
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG11 CM2880 G106	4332695
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG11 CM2880 G108	4332901
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG12 CM2880 G107	4332690
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG12 CM2880 G109	4332906
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/336/336-019-031.html)	ISL8.9 CM2880 L112	4358493
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/379/379-019-031.html)	ISB/ISD6.7 CM2880 B126	4383693
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/383/383-019-031-tr.html)	ISB5.9 CM2880 B127	4383645

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/389/389-019-031-tr.html)	QSF3.8 CM2880 F112	4383825
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/403/403-019-031.html)	QSL9.3 CM2880 L113	4383811
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/387/387-019-031.html)	ISF3.8 CM2220 F116	4383664
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/436/436-019-031.html)	QSC8.3 CM2880 C102	4388785
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/451/451-019-031.html)	QSB5.9 CM2880 B139	4388870
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/449/449-019-031.html)	QSB3.9 CM2880 B138	5411050
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/506/506-019-031.html)	ISD6.7 CM2880 D101	5411372
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/497/497-019-031.html)	ISF2.8 CM2220 F129	5411325

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Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/501/501-019-031.html)	ISF4.5 CM2220 F123	5411320
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/417/417-019-031.html)	QSG12 CM2880 G112	4388731
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/556/556-019-031.html)	ISF3.8 CM2220 F134B	5504165
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/218/218-019-043.html)	QSB4.5 CM2150 B108	4326163
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/216/216-019-043.html)	QSB6.7 CM2150 B109	4326168
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/331/331-019-043.html)	QSB7 CM2880 B117	4358390
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/284/284-019-043.html)	QSB2.8 CM2880 F104	4332741
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/284/284-019-043.html)	QSB2.8 CM2880 F108	4332746

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Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/209/209-019-043-tr.html)	ISB3.9 CM2200 B107	4310792
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/164/164-019-043.html)	ISL9.5 CM2150 SN	4310608
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/206/206-019-043-tr.html)	QSNT14 CM876 N102	4325993
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/177/177-019-043-tr.html)	ISG11 CM2880 G106	4332695
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/325/325-019-043-tr.html)	ISG11 CM2880 G108	4332901
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/177/177-019-043-tr.html)	ISG12 CM2880 G107	4332690
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/325/325-019-043-tr.html)	ISG12 CM2880 G109	4332906
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/336/336-019-043.html)	ISL8.9 CM2880 L112	4358493

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Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/383/383-019-043-tr.html)	ISB5.9 CM2880 B127	4383645
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/389/389-019-043-tr.html)	QSF3.8 CM2880 F112	4383825
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/403/403-019-043.html)	QSL9.3 CM2880 L113	4383811
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/387/387-019-043.html)	ISF3.8 CM2220 F116	4383664
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/436/436-019-043.html)	QSC8.3 CM2880 C102	4388785
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/451/451-019-043.html)	QSB5.9 CM2880 B139	4388870
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/449/449-019-043.html)	QSB3.9 CM2880 B138	5411050

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Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/506/506-019-043.html)	ISD6.7 CM2880 D101	5411372
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/497/497-019-043.html)	ISF2.8 CM2220 F129	5411325
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/501/501-019-043.html)	ISF4.5 CM2220 F123	5411320
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/417/417-019-043.html)	QSG12 CM2880 G112	4388731
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/556/556-019-043.html)	ISF3.8 CM2220 F134B	5504165